Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Wilcox, William R Ph.D.		
Project Title:	Residual Gas Effects on Detached Solidification in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	13699-5705	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Regel, Liya L. (Clarkson University)		
Grant/Contract No.:	NAG8-1703		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A system was perfected that controls the oxygen concentration over a sessile drop of molten semiconductor InSb while permitting the surface on which the drop rests to be tilted. The drop shape was measured while the surface was horizontal in order to give the surface tension of the melt in addition to the static contact angle. Contact angle hysteresis was measured by slowly tilting the surface until the drop began to move. Our long-term goal was to make detached solidification reproducible, which requires a full understanding of the mechanisms underlying it. Our Moving Meniscus Model of steady-state detachment predicted that it depends strongly on the surface tension of the melt and the advancing contact angle with the ampoule wall. The objective of the current task was to determine the influence of residual gases on the surface tension and contact angle of molten semiconductors on typical ampoule materials. Our focus was on the influence of oxygen on indium antimonide (InSb) on clean fused silica (“quartz”). Greatly superior crystals are produced when detached solidification is achieved, with orders of magnitude fewer dislocations, grain boundaries and twins. Some behavior observed during detached solidification has been attributed to the presence of residual gases in the ampoule,		

	particularly oxygen and water vapor. However, there were no previous data on the influence of these gases on the surface tension or contact angle of molten semiconductors, except some excellent Japanese results for the influence of oxygen on the surface tension of molten silicon.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We now understand how to produce superior crystals by detached solidification not only in space but also on Earth. Furthermore, we also now understand the interaction between oxygen, other gases, and the semiconductor melt. This is helpful in growing semiconductors by a variety of techniques.
Task Progress:	This project officially expired on November 30, 2004. However, the remaining graduate student will not complete the research until the summer of 2005. We have overcome great difficulties in developing a system to reach very low oxygen concentrations in a flowing gas system, rotating the plate on which the sessile drop rests, obtaining a high quality silhouette image of the drop, converting the drop profile to x-y values, and using these values to obtain precise reproducible values for the left and right contact angles. Our research has shown that oxygen does not increase the contact angle of molten InSb on fused silica (as had been hypothesized in the literature). Instead, oxygen appears to lower the contact angle because of adsorption of dissociated oxygen molecules on the surface of the molten drop.
Bibliography Type:	Description: (Last Updated: 03/03/2005)
Articles in Peer-reviewed Journals	Wang, J.; Regel, L.L.; Wilcox, W.R. "Detached solidification of InSb on Earth" Journal of Crystal Growth, vol 260, pp 599 , Mar-2004